

packages/graph-view/src/utils/containers.ts

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Overview

A new utility module `packages/graph-view/src/utils/containers.ts` is added. It derives container metadata for graph nodes by grouping them either by folder structure or, when necessary, by community detection.

Key changes

- **Imports** (lines 1-5): `GraphNode`, `GraphEdge` from `@novadiff/graph-core/types` and `detectCommunities` from `./louvain`.
- **Interfaces** (lines 7-12, 14-17): `DerivedContainer` and `DeriveResult` expose container id, name, node ids, and strategy.
- **Folder grouping** (lines 52-75): `groupByFolder` uses `commonPrefix` (lines 30-45) and `firstSegment` (lines 47-51) to bucket nodes by the first path segment after trimming the longest common prefix.
- **Fallback decision** (lines 77-89): `shouldFallbackToCommunity` returns true when bucket count is below `MIN_BUCKET_COUNT` or any bucket exceeds `MAX_CONCENTRATION`.
- **Derivation entry point** (lines 91-156): `deriveContainers` orchestrates grouping, optional community fallback, suppression of single-child containers (when `nodes.length >= MIN_NODES_FOR_SUPPRESSION`), and returns `{ containers, ungrouped }`.
- **Constants** (lines 19-23): `MIN_BUCKET_COUNT`, `MAX_CONCENTRATION`, `MIN_NODES_FOR_SUPPRESSION`, `ROOT_BUCKET` control the heuristics.

Impact

- **Deterministic grouping:** Nodes are consistently bucketed by folder or community.
- **Optional community detection:** Invoked only when `shouldFallbackToCommunity` deems folder grouping insufficient, reducing unnecessary analysis.
- **Single-child suppression:** Containers with a single node are removed and the node is returned in `ungrouped`.
- **Root bucket handling:** Nodes without a `filePath` are placed in a special `ROOT_BUCKET` container.
- **No API changes:** Existing consumers can import the new module without altering existing code.

Risks & follow-ups

- **Community detection integration:** The behavior of `detectCommunities` is not shown in the diff; verify that it returns `(nodeId, cid)` pairs as expected.
- **Edge-case handling:** Ensure nodes lacking `filePath` are correctly routed to `ROOT_BUCKET` and that suppression respects `MIN_NODES_FOR_SUPPRESSION`.

- **Threshold tuning:** `MIN_BUCKET_COUNT` and `MAX_CONCENTRATION` may need adjustment for different graph sizes; monitor clustering quality.
- **Performance regression:** Benchmark `deriveContainers` on large graphs to confirm community fallback does not introduce significant overhead.